

Ex 8: Socket Programming using TCP

a)Echo:

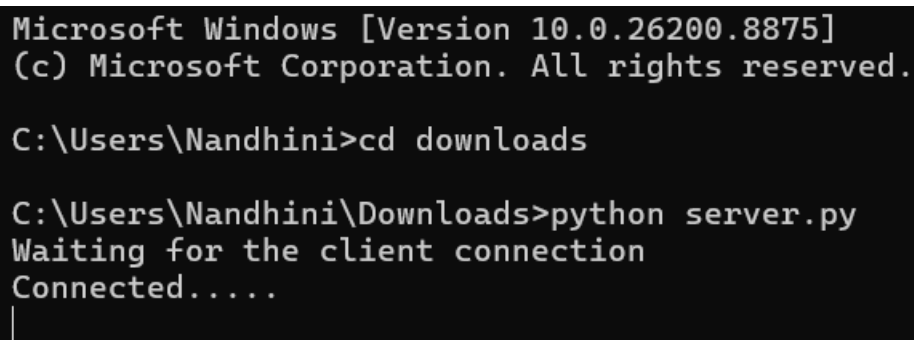
Server Program:

```
import socket
s=socket.socket()
s.bind(('localhost',9937))
s.listen(3)
print("Waiting for the client connection")
c,addr=s.accept()
c.send("Connected successfully".encode())
print("Connected.....")
while True:
    c1=c.recv(1024).decode()
    c.send(c1.encode())
c.close()
S.close()
```

Client Program:

```
import socket
s=socket.socket()
s.connect(('localhost',9937))
print(s.recv(1024).decode())
while True:
    c=input("c1:")
    s.send(c.encode())
    s1=s.recv(1024).decode()
    if s1=="break":
        break
    print("Server:",s1)
S.close()
```

Output for a):



```
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nandhini>cd downloads

C:\Users\Nandhini\Downloads>python server.py
Waiting for the client connection
Connected.....
|
```

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Nandhini> cd downloads
PS C:\Users\Nandhini\downloads> python client.py
Connected successfully
cl:hello
Server: hello
cl:hii
Server: hii
cl:hihi i'm nandhini
Server: hihi i'm nandhini
cl:i'm okay
Server: i'm okay
cl:

```

b) Chat:

Server Program:

```

import socket
server = socket.socket()
server.bind(('localhost', 9007))
server.listen(1)
print("Waiting for client...")
client_socket, address = server.accept()
print("Client connected!")
while True:
    msg = client_socket.recv(1024).decode()
    if not msg:
        break
    print("Client:", msg)
    reply = input("You: ")
    client_socket.send(reply.encode())
client_socket.close()
server.close()
print("Chat ended.")

```

Client Program:

```

import socket
client = socket.socket()
client.connect(('localhost', 9007))
print("Connected to server!")
while True:
    msg = input("You: ")
    client.send(msg.encode())
    reply = client.recv(1024).decode()
    if not reply:
        break
    print("Server:", reply)
client.close()

```

```
print("Chat ended.")
```

Output for b)

```
Waiting for client...
Client connected!
Client: hii
You: hello
Client: how are you?
You: I'm fine
Client: glad to hear it
You: |
```

```
PS C:\Users\Nandhini\downloads> python chatclient.py
Connected to server!
You: hii
Server: hello
You: how are you?
Server: I'm fine
You: glad to hear it
|
```

c) File Transfer:

Server Program:

```
import socket
s=socket.socket()
s.bind(('localhost',9004))
s.listen(3)
print("Waiting for the client connection")
c,addr=s.accept()
c.send("Connection successfully".encode())
print("Connected...")
filename=c.recv(1024).decode()
file1=open(filename,"r")
data=file1.read()
file1.close()
file2=open("Server_"+filename,"w")
file2.write(data)
file2.close()
c.send("File read and written successfully".encode())
c.close()
s.close()
```

Client Program:

```
import socket
s=socket.socket()
s.connect(('localhost',9004))
print(s.recv(1024).decode())
```

```
filename=input("Enter filename to send/read on server:")
s.send(filename.encode())
print(s.recv(1024).decode())
s.close()
```

Output for C):

```
PS C:\Users\Nandhini\downloads> python fileserver.py
Waiting for the client connection
Connected...
PS C:\Users\Nandhini\downloads> |
```

```
PS C:\Users\Nandhini> cd downloads
PS C:\Users\Nandhini\downloads> python fileclient.py
Connection successfully
Enter filename to send/read on server:hello.txt
File read and written successfully
PS C:\Users\Nandhini\downloads> |
```

 hello.txt	29-07-2026 20:17	Text Document	1 KB
 Server_hello.txt	29-07-2026 20:17	Text Document	1 KB